الصناعات الدقيقة
PRECISION INDUSTRIES®
Power Generator Packagers And Canopy Builders

# GENSET NO LOAD – LOAD & PRE-TEST CHECKING REPORT

| | | | |
|---|---|---|---|
| Date : | 12 / JULY / 2017 | | |
| PI Model : PI 313C | | Type : | CLOSE |
| Prime Power : 284 KVA | | Canopy Color : | ORANGE |
| Frq. : 60 Hz | $P.F_{LOAD}$ 1 | Voltage : | 380 V |
| Serial No : | | | |
| Job No : | 8650 | | |

CLIENT : SAUDI ARBIAN SERVICE

PROJECT :

| GENSET DETAILS : | | | | | | | |
|---|---|---|---|---|---|---|---|
| Engine : | CUMMINS | | | Model No : | QSL9 – G3 | Serial No : | 84507969 |
| Alternator : | STAMFORD | | | Model No : | UCDI 1274K1 | Serial No : | N17D166100 |
| Controller : | DEEP SEA | | | Model No : | 6120 | Serial No : | 6419529 |
| C. Breaker : | METASOL | Amps | 400 A | Model No : | ABN 403C | Serial No : | N/A |
| C. Transformer : | ELEMEX | Amps | 800/5 A | Model No : | N/A | Serial No : | N/A |
| AVR : | STAMFORD | | | Model No : | SX460 | Serial No : | 111505 - 3287 |

| GENSET RATING : | | | |
|---|---|---|---|
| Testing Full Load Current : | 345 Amps | Nominal Current : | 432 Amps |

| LOAD TEST WILL BE IMPLEMENTED AL AMIENT CONDITIONS | | | |
|---|---|---|---|
| Temperature : | $40^{0}$ C | Altitude : | |
| | | | |

LOAD TEST

| LOAD | RUN TIME | FREQ | SPEED | VOLTAGE L-L | | | CURRENT | | | POWER | COOLANT TEMP. | OIL PRESSURE | CHARGING ALTERNATOR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | L1 | L2 | L3 | L1 | L2 | L3 | | | | |
| % | Min | Hz | RPM | V | V | V | A | A | A | KW | $^{o}C$ | BAR | VDC |
| 0% | 5 | 60.0 | 1800 | 380 | 380 | 380 | -- | -- | -- | -- | 74 | 5.04 | 28.4 |
| 24% | 5 | 60.0 | 1800 | 380 | 381 | 381 | 83 | 84 | 84 | 55 | 80 | 4.76 | 28.4 |
| 51% | 5 | 60.0 | 1800 | 380 | 382 | 382 | 175 | 175 | 174 | 115 | 81 | 4.52 | 28.4 |
| 75% | 5 | 60.0 | 1800 | 381 | 382 | 382 | 260 | 260 | 260 | 171 | 84 | 4.36 | 28.6 |
| 100% | 10 | 60.0 | 1800 | 381 | 382 | 382 | 345 | 346 | 345 | 225 | 90 | 4.16 | 28.6 |
| 110% | 10 | 60.0 | 1800 | 381 | 382 | 382 | 380 | 381 | 380 | 250 | 92 | 4.08 | 28.6 |

LOAD ACCEPTANCE TEST:

| SUDDENLY BLOCK LOAD | VOLTAGE | FREQUENCY | CURRENT | RESPONSE TIME |
|---|---|---|---|---|
| % | V | Hz | A | SEC |
| 51% | 380 | 60.0 | 175 | 0 |
| 0% | 389 – 380 | 61.1 – 60.0 | 0 | 5 |
| 52% | 374 – 380 | 59.2 – 60.0 | 181 | 5 |

PROTECTION SYSTEM TEST:

| OVER FREQUENCY SD | UNDER FREQUENCY SD | OVER VOLTAGE SD | UNDER VOLTAGE SD | OIL PRESSURE SD | COOLANT LEVEL SD | COOLANT TEMP SD |
|---|---|---|---|---|---|---|
| 67 | 53 | 460 | 340 | 1.0 | OK | ECM 105 |

| ACCESSORIES DETAILS : | | | | |
|---|---|---|---|---|
| | Model | | Serial No | |
| | Model | | Serial No | |
| | Model | | Serial No | |

| Test Carried out by PI Supervisor : | Witnessed and approved by Client : |
|---|---|
| Name : Eng. FARHAN SALEEM | Name : |
| Signature : | Signature : |
| Date : 12 / JULY / 2017 | Date : |

PI PRECISION INDUSTRIES
TESTED
Testing Department

PO BOX NO. 17, BARNACK ROAD, STAMFORD LINCOLNSHIRE, PE9, 2NB, ENGLAND.

STAMFORD

# Test Certificate

FOR

SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR

Machine No. **N17D166100**

**Type Test has been conducted on similar design machine**
STANDARDS: **Generator generally conforms to BS 5000 : Part 3,**
**Tested in accordance with BS EN 60034-1/IS/IEC 60034-1**
**Rotor dynamically balanced to BS ISO 1940 - 1**

**ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:**

| | | | | |
|---|---|---|---|---|
| FRAME: UCDI274K1 | **AVR:** SX460 | **AVR NO:** 1115053281 | **ENCLOSURE:** IP23 | |
| RATED KVA: 250 | **RATED VOLTS:** 400 | **RATED AMPS:** 360.8 | **FREQUENCY:** 50 | **PHASE:** 3 |
| POWER FACTOR: .8 | **STR CONN:** Series Star | **STR WINDING:** 311 | **DUTY:** Base Continuous | **WIRE:** 4-Wire |
| EXC. VOLTS: 79 | **EXC.AMPS:** 2.9 | **AMB TEMP:** 40 | | |

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

| WINDING | COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES) | INSULATION RESISTANCE (M- Ohm) | HV TEST 2KV FOR 1 MIN | INSULATION CLASS |
|---|---|---|---|---|
| STATOR | 0.0267 | 2+ | OK | H |
| ROTOR | 2.164 | 2+ | OK | H |

**REGULATION TEST**

| FREQ. | VOLTS | AMPS | KVA | PF | KW | EXC.VOLT | EXC.AMP | % LOAD |
|---|---|---|---|---|---|---|---|---|
| 52 | 401.35 | 0.00 | 0.00 | 0.00 | 0.00 | 12.3 | .51 | 0.00 |
| 50 | 400 | 360.8 | 250 | .8 | 200 | 79 | 2.9 | 100.00 |

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.UCD27.4.K.1.311.,1,.,14,.SX460.Black - RAL9011
Remarks:

| | | |
|---|---|---|
| Date: 18-APR-17 | Job No: 3481897-001 | AUTHORISED BY<br>**Manoj K Upadhyaya**<br>Quality Assurance |

## Certificate Of Conformity

Quality System Approval
ISO 9001/2008

Frame Size : **UCDI274K1**

Serial Number : **N17D166100** **Manufactured Date :** 18-APR-17

KVA: **250**

Authorized By Quality Assurance: **Manoj K Upadhyaya**

This document is electronically generated **Print Date: 18-APR-17**

# EU DECLARATION OF CONFORMITY

Generator Technologies

This synchronous A.C. generator is designed for incorporation into an electricity generating-set and fulfils all the relevant provisions of the following EU Directive(s) when installed in accordance with the installation instructions contained in the product documentation:

| | |
|---|---|
| 2014/35/EU | Low Voltage Directive |
| 2014/30/EU | The Electromagnetic Compatibility (EMC) Directive |

and that the standards and/or technical specifications referenced below have been applied:

| | |
|---|---|
| EN 61000-6-2:2005 | Electromagnetic compatibility (EMC). Generic standards - Part 6-2: Immunity for industrial environments |
| EN 61000-6-4:2007+A1:2011 | Electromagnetic compatibility (EMC). Generic standards - Part 6-4: Emission standard for industrial environments |
| EN ISO 12100:2010 | Safety of machinery - General principles for design - Risk assessment and risk reduction |
| EN 60034-1:2010 | Rotating electrical machines - Part 1: Rating and performance |
| BS ISO 8528-3:2005 | Reciprocating internal combustion engine driven alternating current generating sets - Part 3: Alternating current generators for generating sets |
| BS 5000-3:2006 | Rotating electrical machines of particular types or for particular applications - Part 3: Generators to be driven by reciprocating internal combustion engines - Requirements for resistance to vibration |

This declaration has been issued under the sole responsibility of the manufacturer. The object of this Declaration is in conformity with the relevant Union harmonization Legislation.

The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Generator Technologies Limited, 49/51 Gresham Road, Staines, Middlesex, TW18 2BD, U.K.

| | |
|---|---|
| Signed:<br><br>Date: 01st February 2016 | Name, Title and Address:<br>Kevan J Simon<br>Global Technical and Quality Director<br>Cummins Generator Technologies<br>Fountain Court<br>Lynch Wood<br>Peterborough, UK<br>PE2 6FZ |

Description UCDI274K1

Serial Number N17D166100

450-16383-F